VERSION 1 · APRIL 2026

XPLOORIA

Gamified Tourism Discovery & Community Platform

Northeast India · iOS & Android

PLATFORM	BACKEND	COMPANY
Flutter (iOS + Android)	Firebase (spotmizoram)	HillsTech
FEATURES	**STATE MANAGEMENT**	**VERSION**
61+ Screens · 50+ Routes	Riverpod 3	1.0 Release

XPLOORIA — Version 1 Product Documentation

Gamified Tourism Discovery & Community Platform for Northeast India

Platform: iOS & Android (Flutter)

Backend: Firebase (Firestore, Auth, Storage, Analytics)

Version: 1.0

Build Date: April 2026

Firebase Project: `spotmizoram`

Bundle ID (iOS): `com.hillstech.spotmizoram`

Company: HillsTech

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1. Executive Summary

Xplooria is a gamified, community-driven tourism discovery mobile application purpose-built for Northeast India, starting with Mizoram. It solves the critical gap between the extraordinary natural and cultural richness of Northeast India and its near-total absence from mainstream digital travel platforms.

Version 1 ships a fully functioning dual-facing platform:

- **Consumer app** — Discover places, plan trips, earn XP, build social travel lists, compete on leaderboards, and book curated tour packages.
- **Admin panel** — Embedded within the same app binary; accessible to super-admin accounts only; provides full content management, analytics, moderation, and booking operations.

The platform is engineered on Flutter + Firebase, enabling a single codebase to target both iOS and Android with real-time data, atomic gamification transactions, and role-based access control.

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2. Product Vision & Mission

Vision

> *To be the definitive digital guide and social layer for tourism across all eight states of Northeast India — making the region's hidden gems discoverable, bookable, and shareable by the world.*

Mission

- Make Mizoram's landscapes, culture, and cuisine digitally discoverable and accessible to domestic and international travellers.

- Incentivise locals and travellers to contribute, review, and share content through a fair, transparent XP-and-badge reward system.
- Provide local tour operators a no-cost digital storefront to list, promote, and take bookings for their packages.
- Build a self-sustaining community of travellers and content contributors whose engagement continuously enriches the platform.
- Expand across all eight Northeast Indian states as version milestones are reached.

Tagline Options Tested

- *"Discover the East"*
- *"Where every journey earns a badge"*
- *"Northeast India's Travel Layer"*

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3. Core Problems Solved

PROBLEM	HOW XPLOORIA SOLVES IT
Northeast India is invisible on global travel apps	Dedicated, curated, locally-sourced database of 8+ place categories
Travellers have no reliable local knowledge source	Community reviews, photos, and contributions from verified locals
Tour operators have zero digital presence	Free venture/package listing with booking management
Passive content consumers give no feedback	Gamification — every review, photo, and contribution earns XP and badges
No social layer for group travel planning	Bucket lists (collaborative rooms), Dares (group challenges), Dilemmas (A/B travel polls)
Language/knowledge barriers for first-time visitors	State-by-state Visitor Guides with Dos/Don'ts, Quick Facts, local etiquette
No single place to find events	Events section with calendar view, ticketing metadata, and filtering

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4. Platform Architecture

4.1 Layered Architecture Overview

```
┌──────────────────────────────────────────────────────────────────┐
│  PRESENTATION LAYER (Flutter Widgets & Screens)                  │
│  61+ Screen files | 3 Widget libraries | Material 3 Dark-first   │
├──────────────────────────────────────────────────────────────────┤
│  STATE MANAGEMENT LAYER (Riverpod 3)                             │
│  14 Controllers | AsyncNotifiers | StreamProviders               │
├──────────────────────────────────────────────────────────────────┤
│  SERVICE LAYER (Firestore + HTTP)                                │
│  30+ Firestore services | Dio HTTP client | Firebase Storage     │
├──────────────────────────────────────────────────────────────────┤
│  DATA LAYER (Freezed Models)                                     │
│  13 Model files | Immutable classes | JSON serialization         │
├──────────────────────────────────────────────────────────────────┤
│  BACKEND (Firebase Cloud)                                        │
│  Firestore 30+ collections | Auth (custom claims) | Storage      │
│  Analytics | Real-time Database (LatLng lookups)                 │
└──────────────────────────────────────────────────────────────────┘
```

4.2 State Management Pattern

All state is managed by **Riverpod 3** using:

- `AsyncNotifier` — for async CRUD operations (auth, gamification, listing management)
- `Notifier` — for synchronous local state (pagination, dare list, bucket list)
- `StreamProvider` — for real-time Firestore subscriptions (community posts, bookings, banners)
- `FutureProvider` — for one-time data fetches with `.autoDispose` for admin collections

4.3 Data Flow (Gamification Example)

```
User submits a review
       ↓
place_detail_screen → GamificationController.award(XpAction.writeReview)
       ↓
GamificationService.award() — Firestore TRANSACTION:
  ① Read users/{uid}
```

```
  ② Parse StreakInfo (streak, multiplier 1.0–2.0)
  ③ Calculate xpAwarded = baseXp(15) × multiplier
  ④ Evaluate 22 badges → collect newBadgeIds[]
  ⑤ Recalculate level (1–10)
  ⑥ UPDATE users/{uid} (points, level, badges, streak, lastLogin)
  ⑦ WRITE xpEvents sub-doc (audit log)
        ↓
GamificationResult pushed to broadcast stream
        ↓
XpToastOverlay shows animated "+15 XP" toast (3.5 seconds)
        ↓
authController.refreshProfile() → all profile UI rebuilds
```

4.4 Router Architecture

Navigation is handled by **GoRouter 17** with a `_RouterNotifier` (singleton `ChangeNotifier`) that:

1. Caches the latest auth and admin state to avoid GoRouter recreation on every provider change.
2. Implements redirect guards for three tiers: unauthenticated, authenticated, and super-admin.
3. Uses `HeroControllerScope` to isolate admin navigation animations from the main shell.

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5. Technology Stack

5.1 Core Framework

PACKAGE	VERSION	PURPOSE
Flutter SDK	≥ 3.10.8	Cross-platform mobile UI
Dart SDK	≥ 3.0	Language
flutter_riverpod	^3.1.0	Reactive state management
go_router	^17.1.0	Type-safe declarative navigation

5.2 Firebase Services

SERVICE	USAGE
Firebase Auth	Email/password authentication; custom claims for admin role
Cloud Firestore	Primary database; 30+ collections; real-time streams
Firebase Storage	Image uploads (profile, listing, dare, guide)
Firebase Analytics	User event tracking
Firebase Realtime DB	Geo/LatLng data lookup

5.3 UI & UX Libraries

PACKAGE	PURPOSE
iconsax	Iconsax icon set (100+ icons used throughout)
google_fonts (Inter)	Typography
cached_network_image	Lazy image loading with memory/disk cache
carousel_slider	Home banner carousel, onboarding
smooth_page_indicator	Carousel dot indicators
shimmer	Loading skeleton states
lottie	JSON animation (achievement, empty states)
flutter_animate	Micro-animations
fl_chart	Admin analytics charts
confetti	Achievement celebration animations
percent_indicator	XP progress bars

5.4 Maps & Location

PACKAGE	PURPOSE
flutter_map	OpenStreetMap rendering (Community map tab)
latlong2	Coordinate model (LatLng)
geolocator	GPS positioning
geocoding	Address ↔ GPS conversion

5.5 Media & Forms

PACKAGE	PURPOSE
image_picker	Camera / gallery photo selection
image_cropper	Pre-upload image cropping
flutter_form_builder	Form field management
form_builder_validators	Built-in validation rules
file_picker	CSV file selection for admin bulk upload
csv	CSV parsing (admin batch import)

5.6 Networking & Storage

PACKAGE	PURPOSE
dio	HTTP client with interceptors
flutter_secure_storage	Encrypted local credential storage
shared_preferences	Key-value settings (theme mode, etc.)
path_provider	File system path resolution

5.7 Utilities

PACKAGE	PURPOSE
intl	Date/time formatting
timeago	Relative timestamps ("2 hours ago")
uuid	Unique ID generation
url_launcher	External URL and deep-link opening
share_plus	System share sheet
connectivity_plus	Network reachability detection
dartz	Functional Either/Option error handling

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6. User Roles, Functions & Duties

Xplooria has three distinct user roles, each with different capabilities and access levels.

6.1 Guest / Unauthenticated User

Who they are: Any visitor who opens the app without logging in.

What they can do:
- Browse all publicly listed tourist spots, restaurants, cafes, hotels, homestays, adventure spots, shopping areas, and events.
- View spot details (images, descriptions, opening hours, entry fees, facilities, reviews).
- Browse Tour Ventures (packages) and view details.
- View the Community map with all pins.
- Read reviews and community posts.
- Browse the State Visitor Guides.
- View the Leaderboard.

What they cannot do:
- Write reviews or upload photos.
- Create community posts, bucket lists, dilemmas, or dares.
- Book ventures.
- Earn XP or badges.
- Access the Admin panel.

Impact: Guests represent the top of the funnel. The richness of public content is the primary conversion lever to registration.

6.2 Registered User

Who they are: Any user who has created an account (email + password).

Permissions: All guest permissions PLUS the following active participation capabilities.

Core Duties & Capabilities:

AREA	CAPABILITY
Profile	Create and edit profile (name, bio, avatar, location)
Reviews	Write a review and rate (1–5 stars) any spot or listing (one per place per user)
Photos	Upload community photos to any listing
Content Contribution	Submit new spots/places for admin review and approval
Bookmarks	Save places to personal bookmarks
Events	View and participate in events
Booking	Book tour ventures with selected packages and add-ons
Gamification	Earn XP, level up (1→10), unlock 22 badges, appear on leaderboard
Streaks	Maintain daily login streaks for an XP multiplier bonus

Community Features:

FEATURE	CAPABILITY
Community Feed	Create posts, tips, questions, and photo reports
Dilemmas	Create and vote on A/B travel choice polls
Bucket Lists (Rooms)	Create collaborative destination lists; invite members; tick off items as a group

FEATURE	CAPABILITY
Dares	Create and join group travel challenges with photo-proof submission
Dare Rewards	Scratch-off reward cards earned by completing dare milestones
Notifications	Receive real-time push/in-app alerts for dare join requests, approvals, and submissions

Venture Booking Workflow:

1. Browse ventures on Home screen or Ventures tab.
2. Select a package tier and optionally add add-ons (packed lunch, guide, transport, etc.).
3. Review booking summary (total cost breakdown, operator details).
4. Submit booking → status becomes `pending`.
5. Track booking status in "My Bookings" screen.
6. Post-trip: submit feedback and rating.

6.3 Super Administrator

Who they are: Accounts whose Firebase Auth custom claims include `{ "superAdmin": true }`. Set via the Node.js admin script (`scripts/set_admin_claim.js`).

Access: Separate `/admin` navigation shell within the same app binary. Inaccessible to regular users. The GoRouter redirect enforces this at every navigation attempt.

Core Admin Duties & Capabilities:

CONTENT MANAGEMENT

DUTY	DETAILS
Manage Tourist Spots	Create, edit, delete spots; manage images, categories, facilities, entry fees
Manage Restaurants	Full CRUD for restaurants; price range, cuisine types, delivery flags
Manage Accommodations	Manage hotels + homestays in a merged view
Manage Cafes	Full CRUD for cafes; specialties, opening hours
Manage Adventure Spots	Create/edit adventure categories

DUTY	DETAILS
Manage Shopping Areas	Create/edit shopping venues
Manage Events	Create/edit/delete events with ticketing metadata
Manage Ventures	Full CRUD for tour packages; set operator info, highlights, itinerary challenges, add-ons
Bulk CSV Upload	Import multiple listings at once via CSV file (per collection)
Sort & Filter	Sort by name or newest; toggle grid/list view

BANNER MANAGEMENT

DUTY	DETAILS
Create/Edit Banners	Title, subtitle, image URL, link type (none / external URL / internal route), order
Activate/Deactivate Banners	Toggle `isActive` per banner
Toggle Section Visibility	Global on/off for the entire home banner carousel via `app_config/home_banners`

USER MANAGEMENT

DUTY	DETAILS
View All Users	See full list with XP, level, registration date
Assign/Revoke Roles	Elevate users to admins or revoke access
View User Activity	Review XP history, contributions, bookings, dare participation

VENTURE / BOOKING MANAGEMENT

DUTY	DETAILS
View All Bookings	See every booking request across all ventures

DUTY	DETAILS
Update Booking Status	Confirm, cancel, or mark as completed
Add Admin Notes	Leave internal notes on bookings
View Feedback	Read post-trip user feedback per venture
Packages Tab	See all venture packages and their registration counts

ANALYTICS DASHBOARD

METRIC	DETAILS
Total Users	All registered users count
New Users Today / This Week	New registrations trend
Total Spots & Listings	Per-collection counts
Total Events & Ventures	Content inventory
Community Activity	Posts, reviews, dilemma votes
Booking Statistics	Pending, confirmed, completed
XP Awarded	Total platform gamification output
Charts	Daily active usage trends, submissions by category (via fl_chart)

VISITOR GUIDE MANAGEMENT

DUTY	DETAILS
Create/Edit State Guides	Per-state guide with tagline, about, quick facts, Dos/Don'ts
Publish/Unpublish	Toggle guide visibility per state
Upload Banner Images	State-specific guide hero images

MODERATION

DUTY	DETAILS
Review Contributions	Approve or reject user-submitted spots
Flag Content	Remove posts, reviews, or photos violating guidelines
Bulk Approve/Reject	Process contribution queues efficiently

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7. Feature Deep-Dive: Guest / Public User

7.1 Home Screen

The Home screen is the primary entry point. It is built as a `CustomScrollView` with `SliverAppBar` (floating) and multiple stacked sections:

Sections (top to bottom):

1. **App Bar** — Xplooria logo and notification bell (dare requests badge if signed in).
2. **State Selector** — Subtitle "Exploring [Mizoram]" is tappable; opens a bottom sheet picker for 8 NE states (only Mizoram active in V1; others show "Coming Soon").
3. **Home Banner Carousel** — Firestore-driven promotional banners with links (internal routes or external URLs). Hidden when admin disables the section. Smooth dot indicators.
4. **Browse Listings Grid** — 4-column grid of category shortcuts: Tourist Spots, Restaurants, Stay, Cafes, Adventure, Shopping, Events, Community. Each taps to the respective Listings tab.
5. **Visitor Guide Card** — Tappable state guide card for the selected NE state; opens full guide.
6. **Featured Spots Section** — Horizontally scrollable cards with category filter tabs (All, Mountains, Waterfalls, Cultural Sites, Viewpoints, Adventure). Cards show image, category, name, location, rating, and views.
7. **Tour Packages Section** — Horizontally scrollable venture cards from the live `ventures` collection. Shows image, category chip, difficulty chip, duration, price, location, and operator.
8. **AI Travelling Planner (Locked)** — "Coming Soon" teaser section with locked badge, feature chips (Smart Itineraries, AI Companion Chat, Personalised Tips, Live Suggestions), and launch announcement.
9. **Quick Stats (Signed-in only)** — XP, contributions count, badges count stat cards.

7.2 Spots Screen

- Paginated list of tourist spots pulled from `spots` Firestore collection.

- Filter chips for 6 categories (Mountains, Waterfalls, Cultural Sites, Viewpoints, Adventure, Lakes/Caves).
- Each card shows hero image, category, name, district, rating, and view count.
- Tapping opens `SpotDetailScreen`: full image gallery, description, facilities, entry fees, "Things to Do" list, add-ons, and review thread.

7.3 Listings Screen

Tabbed screen for 7 listing types:

TAB	FIRESTORE COLLECTION	EXTRA DETAILS
Restaurants	`restaurants`	Cuisine types, price range, delivery/reservation flags
Stay	`accommodations` + `homestays`	Merged view of hotels and homestays
Cafes	`cafes`	Specialties, opening hours
Adventure	`adventureSpots`	Difficulty, group info
Shopping	`shoppingAreas`	Category, map
Events	`events`	Dates, ticketing, type
Ventures	`ventures`	Full enterprise tour packages

Each listing taps to a full detail screen with gallery, details, map, and reviews.

7.4 Search Screen

- Global search across all collections simultaneously.
- Category filter chips to scope search.
- Real-time debounced Firestore queries.

7.5 Community — Map Tab (Feed)

The Community screen's default "Feed" tab renders a full-screen interactive map powered by `flutter_map` (OpenStreetMap):

- **Map Tiles**: Carto Dark (dark mode) / Carto Light (light mode) adapts to the app theme.
- **Pins**: Two pin types:

- **Animated Pulsing Pins** (tourist spots) — colour-coded by category (Mountains=blue, Waterfalls=cyan, Cultural=amber, etc.); 1.8s pulsing animation.
- **Circular Image Pins** (restaurants, cafes) — circular cropped photo with pin-tip below; border colour indicates category.
- **Filter Bar**: Icon-based filter chips (Iconsax icons) for All, Tourist Spots, Restaurants, Cafes, Adventure, Homestays, Shopping, Events. Chips are pill-shaped with primary colour fill when selected.
- **Search Bar**: Text search that filters visible pins by name, location, or category in real-time.
- **Zoom Controls**: +/- buttons top-right.
- **Tap to Detail**: Tapping any pin opens a bottom sheet (`place_detail_sheet` or `spot_detail_sheet`) with name, category, rating, and quick-navigate CTA.
- **Swipe disabled** on map tab (prevents pan/zoom gesture conflict) via `NeverScrollableScrollPhysics`.

7.6 Leaderboard

- Ranked list of users by total XP.
- Shows rank, avatar, display name, level badge, XP total.
- Top 3 users are visually prominent (gold/silver/bronze styling).

7.7 Visitor Guide

- Per-state digital travel guide.
- Sections: Banner image, tagline, "About the State", Quick Facts (capital, language, population, best season, etc.), Dos, Don'ts.
- Only Mizoram is active in V1. Other states show "Coming Soon" modal when selected.

7.8 Events

- Calendar and list view of events.
- Filters by type and upcoming/past.
- Event detail: title, description, date/time range, location, ticket availability, ticket price.

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8. Feature Deep-Dive: Registered User

All guest features apply, plus:

8.1 Authentication Flow

Register:

1. Enter name, email, password.
2. Firebase Auth creates user.
3. Firestore `users/{uid}` document created with defaults: `points=0, level=1, levelTitle="Explorer"`.
4. Redirected to Home.

Sign In:

1. Enter email and password.
2. Firebase Auth validates.
3. Custom claim checked (`superAdmin`) — regular users go to `/`, admins to `/admin`.

Forgot Password: Firebase Auth sends reset email.

8.2 Profile Screen

Displays:

- Avatar, display name, level badge, XP progress bar (level min → max), level title.
- Stats row: total XP, reviews count, contributions count, photos uploaded.
- Streak banner (🔥 current streak, longest streak).
- Badges grid — earned badges with rarity colour coding (Common/Rare/Epic/Legendary).
- XP History — scrollable activity feed (earned XP event log).
- "My Bookings" shortcut — links to venture bookings.
- "Dare Dashboard" — summary of joined dares and progress.

XP Perks Sheet (tappable XP chip):

- Current level + XP display.
- MezoPerks section (loyalty perks — roadmap in V1, "Coming Soon" pill).
- Recent XP earnings log.

8.3 Writing Reviews

- One review per user per listing (enforced via deterministic review ID: `{userId}_{placeId}`).
- 1–5 star rating + text comment.
- Review atomic-updates the running average rating on the parent listing document.
- Recorded in the global `global_reviews` collection for cross-collection review aggregation.
- Awards **+15 XP** (`XpAction.writeReview`).

8.4 Uploading Community Photos

- Multi-image picker (camera or gallery).
- Optional crop step.

- Uploads to Firebase Storage: `/community_photos/{collection}/{placeId}/{userId}_{timestamp}.ext`.
- Awards **+10 XP** (`XpAction.uploadPhoto`).

8.5 Bucket Lists (Collaborative Rooms)

Concept: Shared destination wish-lists that friends or groups can add to and tick off together.

Create a Bucket List:
- Title, description, banner image, category (11 types), visibility (public/private), max members.
- Awards **+20 XP** (`XpAction.createBucketList`).
- Free plan limit: **5 rooms** per user (`kFreeRoomCap = 5`).

Joining a Room:
- Public rooms: join request → host approves/declines.
- Private rooms: join via unique 6-character join code.

Inside a Room:
- Member list with roles (host/member).
- Items list with category icons, notes, and "checked" state.
- Any member can tick off items → awards **+10 XP** per check.
- Host can add/remove/edit items.

Feature Unlocks (time-based):
- `kConnectUnlockDays = 10` — After 10 days since joining, unlock Connect features.
- `kPokeUnlockDays = 20` — After 20 days, unlock Poke feature.

8.6 Dares (Group Travel Challenges)

Concept: Group-based challenges where participants complete and prove real-world travel tasks.

Create a Dare:
- Title, description, banner, category (11 types: adventure, food rating, photography, etc.), visibility, max participants, deadline.
- Free plan limit: **5 dares** per user (`kFreeDareCap = 5`).

Adding Challenges to a Dare:
- Each challenge: title, description, type (app listing or custom), points, deadline.
- Admin can set medal tier awards (bronze/silver/gold/platinum/special).

Joining a Dare:
- Public: request to join → creator approves.
- Private: join via code.

Completing Challenges:
- Participants submit photo proof for verification.

- Creator reviews submissions.
- On approval: challenge points awarded, medals unlocked.

Dare Rewards:
- Scratch-off cards awarded for milestones.
- Reward types: XP, medal, badge, XP multiplier, or "nothing" (gamble element).
- Each card has a rarity tier.

Notifications:
- In-app notification bell (badge count on app bar).
- Notification types: join requests, approvals, new submissions.

8.7 Dilemmas (A/B Travel Polls)

Short "would you rather?" style polls for community engagement:

- Title and description.
- Option A vs Option B (with images).
- Voting is optimistic (instant UI update, server sync in background).
- Vote counts update in real-time for all viewers.
- Users can change their vote — previous vote is atomically removed from both options before adding new vote.
- Awards **+25 XP** to creators, **+5 XP** to voters.

8.8 Community Feed

- `PostType` enum: `post`, `review`, `tip`, `question`.
- Posts can include text, multiple images, and optionally link to a spot.
- Like/unlike posts (optimistic).
- Comment on posts.
- Infinite scroll pagination (Riverpod paginated notifier).
- Own posts can be deleted (with optimistic removal and rollback on failure).

8.9 Venture Booking

Booking Flow:
1. View venture detail (title, description, category, difficulty, duration, season, operator info, highlights, images, reviews).
2. Select a package tier from the operator's packages.
3. Set group size, special requests.
4. Choose optional add-ons (e.g., packed lunch +₹250, porter +₹500).
5. Review booking page: full breakdown (base price × people + add-ons = grand total, operator contact).

6. Submit → `bookings/{docId}` created with `status: pending`.
7. Track under "My Bookings" (pending → confirmed → completed).
8. Post-completion: submit rating + written feedback.

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9. Feature Deep-Dive: Administrator (Super Admin)

9.1 Admin Shell & Navigation

The admin area (`/admin`) is a separate `ShellRoute` with its own navigation:

- **`NavigationRail`** (wide screens / tablets) — vertical persistent side navigation.
- **`BottomNavigationBar`** (narrow/phone screens) — 5 tabs: Dashboard, Listings, Users, Analytics, Dares.
- Ventures management is embedded inside the **Listings → Ventures tab**.
- Animated route transitions use `HeroControllerScope` isolated from the main app shell.

9.2 Admin Dashboard

Real-time stats from `AppAnalyticsSnapshot` (Firestore: `app_analytics/daily_snapshot`):

METRIC	DATA SOURCE
Total Users	`users` collection count
New Users Today	Timestamp-filtered
New Users This Week	Timestamp-filtered
Total Spots	`spots` collection count
Total Listings	Sum across restaurants, cafes, hotels, etc.
Total Events	`events` count
Total Ventures	`ventures` count
Total Community Posts	`community_posts` count
Total Reviews	`global_reviews` count
Pending Bookings	`bookings` filtered by `status: pending`

METRIC	DATA SOURCE
Total Bookings	`bookings` total count
Total XP Awarded	Sum of all `xpEvents` values

9.3 Listings Management (Admin)

TabBar with 8 tabs: Spots, Restaurants, Accommodations, Cafes, Adventure, Shopping, Events, Ventures.

Per-tab features:
- Live data via `autoDispose` FutureProvider (fetched fresh on each visit).
- Toggle between **List view** (rows with thumbnail, name, location, edit/delete) and **Grid view** (2-column image cards).
- **Sort** by Name (A–Z) or Newest (by `createdAt` timestamp).
- **Filter bar** collapses on scroll (animates via `AnimatedSize`).
- **Edit** button → navigates to form screen with fields pre-populated.
- **Delete** button → shows confirmation dialog → calls `adminListingNotifier.deleteListing()`.
- **Bulk CSV Upload** (via upload icon) → parses CSV and batch writes to Firestore.
- **FAB** (Floating Action Button) → navigates to add form (hidden on Ventures tab — uses embedded screen's own "New" button).

Ventures tab specifically renders the full `AdminVenturesScreen` embedded within the Listings screen, with three sub-tabs:
- **Packages** — List all ventures with card UI, tap to manage.
- **Feedback** — All post-trip feedback from users per venture.
- **Bookings** — All booking records with status chips and admin note field.

9.4 Listing Forms (Add/Edit)

Generic listing form (`admin_add_listing_screen.dart`):
- Title, Description, Location, Category, District.
- Images (multi-select + Firebase Storage upload).
- Rating (seed value), Price Range ($ → $$$$).
- Category-specific dynamic fields (amenities for hotels, cuisine types for restaurants, etc.).

Venture form (`admin_venture_form_screen.dart`) — 12 sections:
1. Basic Info (title, tagline, description)
2. Category & Difficulty (16 category types, 4 difficulty levels)
3. Season (8 season options)
4. Duration (days, nights)
5. Group & Age (max group size, min age)

6. Booking Policy (advance booking days)
7. Pricing (starting price, currency)
8. Operator Info (name, phone, WhatsApp, email)
9. Location (location text, district)
10. Highlights (add/remove bullet list)
11. Images (hero image URL + image list)
12. Challenges (add/remove venture challenges)

Numeric fields show blank (not "0") for unset values — user-friendly UX for editing.

9.5 Banner Management

- Admin can create banners with: title, subtitle, image URL, link type, link value (internal route or external URL), active flag, and ordering number.
- Global carousel section can be toggled on/off with one tap (writes to `app_config/home_banners`).
- Banners ordered by the `order` integer field.

9.6 Visitor Guide Management

- One guide per NE state (8 states).
- Editable fields: stateName, emoji, tagline, about text, Dos array, Don'ts array, quick facts (label+value+icon), banner image URL, published flag.
- Draft/unpublished guides are not visible in the consumer app.

9.7 Analytics Screen

Powered by `fl_chart`:
- Line charts: daily new users trend.
- Bar charts: bookings by status, XP awarded over time.
- Summary cards for all dashboard metrics.
- Data snapshot from `app_analytics` Firestore document.

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10. Gamification System

Xplooria's gamification engine is a full, production-grade reward system built with atomic Firestore transactions.

10.1 XP Actions & Rewards

ACTION	XP EARNED	TRIGGER
Write a review	+15	Submitting any listing review
Upload a photo	+10	Uploading community photo
Create a bucket list	+20	New bucket list created
Complete a bucket item	+10	Tick off item in a list
Create a dilemma	+25	New A/B poll created
Vote on a dilemma	+5	Casting a vote
Submit a contribution	+20	Submit a new spot for review
Daily login	+5	First app open each calendar day
Streak bonus (7 days)	+10	Awarded at streak milestone
Weekly streak	+30	7-consecutive-day login
Monthly streak	+100	30-consecutive-day login

10.2 Streak Multiplier System

A streak mechanic incentivises daily app engagement:

```
Multiplier = 1.0 + floor(loginStreak ÷ 5) × 0.10
Maximum multiplier = 2.0 (at 50+ day streak)
```

STREAK	MULTIPLIER	BONUS
1–4 days	1.0×	No bonus
5–9 days	1.1×	+10% XP
10–14 days	1.2×	+20% XP
15–19 days	1.3×	+30% XP
20–24 days	1.4×	+40% XP

STREAK	MULTIPLIER	BONUS
25+ days	1.5×+	Increases every 5 days
50+ days	2.0× (cap)	+100% XP

10.3 Level System (10 Levels)

LEVEL	TITLE	XP REQUIRED
1	Explorer	0
2	Wanderer	100
3	Adventurer	250
4	Pathfinder	500
5	Guide	1,000
6	Expert	2,000
7	Master	3,500
8	Legend	5,500
9	Champion	8,500
10	Guardian	12,500

10.4 Badge System (22 Badges)

Exploration Badges (6)

BADGE	TRIGGER
First Review	Submit first review
Ten Reviews	Submit 10 reviews
Fifty Reviews	Submit 50 reviews

BADGE	TRIGGER
First Contribution	Submit first spot
Ten Contributions	Submit 10 spots
25 Contributions	Submit 25 spots

Photography Badges (2)

BADGE	TRIGGER
First Photo	Upload first community photo
Photo Master	Upload 50+ community photos

Streak Badges (3)

BADGE	TRIGGER
3-Day Streak	3 consecutive daily logins
7-Day Streak	7 consecutive daily logins
30-Day Streak	30 consecutive daily logins

Community Badges (2)

BADGE	TRIGGER
First Dilemma	Create first A/B poll
Dilemma Voter	Vote on 50+ dilemmas

Bucket List Badges (2)

BADGE	TRIGGER
Bucket Complete (1)	Complete your first bucket list

BADGE	TRIGGER
Bucket Complete (10)	Complete 10 bucket lists

Special & Achievement Badges (7)

BADGE	TRIGGER	RARITY
Early Adopter	Registered in first month of launch	Legendary
Top 10	Reach global leaderboard top 10	Epic
Level 5	Reach level "Guide"	Rare
Level 10	Reach level "Guardian"	Legendary
Social Butterfly	100+ community post likes received	Rare

Rarity Tiers:
- **Common** (grey) — Easy to obtain, high frequency
- **Rare** (blue #42A5F5) — Moderate effort
- **Epic** (purple #AB47BC) — Significant achievement
- **Legendary** (gold #FFB300) — Exceptional milestone

10.5 XP Reward Notification (Toast)

An animated overlay (`XpToastOverlay`) is positioned above the bottom navigation bar. It:

1. Listens to the `gamificationRewardStreamProvider` (broadcast stream).
2. Slides up from bottom with animation when XP is awarded.
3. Displays: `+{xp} XP`, action label, multiplier badge (if > 1.0×), and level-up celebration.
4. Auto-dismisses after **3.5 seconds**.
5. Shows confetti animation on badge unlock or level-up events.

---

11. Data Models & Firestore Collections

11.1 Firestore Collection Architecture

```
firestore/
├── users/{uid}                      ← User profiles, XP, badges, streaks
│   └── xpEvents/{eventId}           ← XP audit log per user
├── spots/{docId}                    ← Tourist spots
│   ├── reviews/{reviewId}           ← Spot reviews (1 per user)
│   └── communityPhotos/{photoId}    ← Community photo uploads
├── restaurants/{docId}              ← Restaurant listings
│   └── reviews/{reviewId}
├── cafes/{docId}                    ← Cafe listings
│   └── reviews/{reviewId}
├── accommodations/{docId}           ← Hotel listings
│   └── reviews/{reviewId}
├── homestays/{docId}                ← Homestay listings
│   └── reviews/{reviewId}
├── adventureSpots/{docId}           ← Adventure spot listings
│   └── reviews/{reviewId}
├── shoppingAreas/{docId}            ← Shopping area listings
│   └── reviews/{reviewId}
├── events/{docId}                   ← Events with ticketing
├── ventures/{docId}                 ← Tour packages
│   ├── reviews/{reviewId}           ← Venture reviews
│   ├── registrations/{regId}        ← Booking registrations
│   └── feedback/{fbId}             ← Post-trip feedback
├── bookings/{docId}                 ← Top-level booking records
├── community_posts/{docId}          ← Community feed posts
├── dilemmas/{docId}                 ← A/B poll dilemmas
├── bucketLists/{listId}             ← Collaborative travel lists
├── dares/{docId}                    ← Group travel challenges
├── global_reviews/{reviewId}        ← Cross-collection review index
├── home_banners/{bannerId}          ← Home carousel banners
├── visitor_guides/{stateKey}        ← Per-state visitor guides
├── app_admins/{uid}                 ← Admin profiles and permissions
│   └── activityLog/{logId}         ← Admin action audit trail
├── app_analytics/{docId}            ← Platform analytics snapshot
└── app_config/{configKey}           ← Global configuration flags
```

11.2 Key Model Schemas

UserModel — Core user profile stored at `users/{uid}`

```
id, email, displayName, photoURL, bio, location,
role (0=admin, 1=user),
points, level (1–10), levelTitle,
badges[], badgesEarned[],
contributionsCount, ratingsCount, photosCount,
dilemmasCreated, dilemmasVoted,
bucketListsCreated, bucketItemsCompleted,
loginStreak, longestStreak, lastLogin,
bookmarks[], createdAt
```

VentureBooking — Tour booking record at `bookings/{id}`

```
id, userId, userName, userEmail,
ventureId, ventureTitle, heroImage, category, location,
operatorName, operatorPhone, operatorWhatsapp, operatorEmail,
selectedPackageName, pricePerPerson, personCount,
selectedAddons[], addonSubtotal, grandTotal,
status (pending|confirmed|cancelled|completed),
adminNote, hasFeedback, createdAt, updatedAt
```

DareModel — Group challenge at `dares/{id}`

```
id, title, description, bannerUrl,
category (11 types), customCategory,
visibility (public|private), maxParticipants, joinCode,
creatorId, creatorName, creatorPhoto,
challenges[] (DareChallenge),
members[] (DareMember),
createdAt, deadline, xpReward,
requiresProof, tags[], status
```

BucketListModel — Collaborative room at `bucketLists/{id}`

```
id, title, description, bannerUrl,
category (11 types), visibility, maxMembers, joinCode,
hostId, hostName, hostPhoto,
items[] (BucketItem), members[] (BucketMember),
```

```
joinRequests[], createdAt, completedAt,
xpReward, badges, challengeTitle
```

---

12. Security Model

12.1 Role Enforcement

Security is enforced at two levels:

1. **GoRouter (Client-side)**: `_RouterNotifier` redirect logic blocks unauthenticated users from protected routes and blocks non-admin users from `/admin` routes.
2. **Firestore Security Rules (Server-side)**: All operations are validated server-side with three helper functions:

```
function isAuth() { return request.auth != null; }
function isUser(uid) { return isAuth() && request.auth.uid == uid; }
function isSuperAdmin() {
  return isAuth() && request.auth.token.superAdmin == true;
}
```

12.2 Data Access Matrix

COLLECTION	PUBLIC READ	AUTH READ	AUTH WRITE	ADMIN ONLY WRITE
users	No (auth only)	Own doc	Own doc	—
spots, restaurants, cafes, etc.	**Yes**	Yes	Yes	—
reviews (sub-collection)	**Yes**	Yes	Create (own)	—
community_posts	**Yes**	Yes	Create / own delete	delete any
bucketLists	Public only if visibility=public	Member read	Host/member	—

COLLECTION	PUBLIC READ	AUTH READ	AUTH WRITE	ADMIN ONLY WRITE
dares	Public if public	Member read	Creator/member	—
bookings	No	Own booking	Create own	Status updates
ventures	**Yes**	Yes	No	Full CRUD
ventures/registrations	No	Own	Create	Update/delete
ventures/feedback	**Yes**	Yes	Create	—
app_admins	Admin only	Own	No	Full CRUD
app_analytics	No	No	No	Full CRUD
home_banners	**Yes**	Yes	No	Full CRUD
visitor_guides	**Yes**	Yes	No	Full CRUD

12.3 Firebase Storage Rules

PATH	READ	WRITE CONDITIONS
`/community_photos/`	Public	Auth + < 10 MB + image/*
`/avatars/{userId}/`	Public	Auth + own userId + < 5 MB + image/*
`/admin_listings/`	Public	Auth + < 15 MB + image/*
`/dare_banners/`	Public	Auth + < 10 MB + image/*
`/dare_proofs/`	Auth	Auth + < 15 MB + image/*
`/banners/`	Public	Auth + < 10 MB + image/*
`/visitor_guides/`	Public	isSuperAdmin + < 10 MB + image/*

12.4 Custom Claim Setup

Super Admin accounts are created via the Node.js utility:

```
scripts/set_admin_claim.js <serviceAccount.json>

Admin Email:    hillstechadmin@spotsence.com
Custom Claim:   { superAdmin: true }
Firestore Doc:  app_admins/{uid}  (permissions: all flags true)
```

---

13. Navigation & Routing

13.1 Shell Structure

The app has two independent `ShellRoute` shells:

1. **Main Shell** — Bottom navigation with 5 tabs: Home, Explore (Spots), Map (Community), Profile, Leaderboard. Available to all authenticated and guest users.
2. **Admin Shell** — Separate navigation rail/bottom bar for admin-only screens. Isolated `HeroControllerScope`.

13.2 Auth Redirect Logic

```
On every navigation:
1. Auth loading? → hold (return null)
2. Not authenticated + protected route? → /login
3. Authenticated + auth route (/login, /register)? → /
4. Route starts with /admin?
   a. Not authenticated? → /login
   b. Admin loading? → hold
   c. Not superAdmin? → / (home, denied silently)
```

13.3 Route Transition Types

- **Slide Up** — Modal screens (auth, booking flow, creation forms)
- **Fade** — Detail screens (spot, listing, event, venture)
- **Default** — Admin screens (no transition override)

---

14. Design System & Theme

14.1 Colour Palette

Brand Colours (theme-invariant):

TOKEN	HEX	USAGE
`AppColors.primary`	`#00E5A0`	CTAs, highlights, active states, FABs
`AppColors.secondary`	`#6C63FF`	Tags, secondary accents
`AppColors.accent`	`#FFB300`	Gold — XP, badges, stars, rewards
`AppColors.success`	`#22C55E`	Success states, easy difficulty
`AppColors.error`	`#EF4444`	Errors, hard difficulty
`AppColors.warning`	`#F59E0B`	Warnings, moderate difficulty

Adaptive Surfaces (dark/light via `context.col`):

TOKEN	DARK	LIGHT	USAGE
`.bg`	`#0A0E1A`	`#F4F7FF`	App background
`.surface`	`#121827`	`#FFFFFF`	Cards, sheets
`.surfaceElevated`	`#1C2333`	`#F0F4FF`	Dropdowns, elevated
`.border`	`#2A3347`	`#E0E8F4`	Dividers, outlines
`.textPrimary`	`#F0F4FF`	`#1A1F35`	Main text
`.textSecondary`	`#8892A4`	`#5A6480`	Labels, hints
`.textMuted`	`#4A5568`	`#A0AEC0`	Disabled text

14.2 Typography (Google Fonts — Inter)

All text uses the **Inter** variable font:

STYLE	SIZE	WEIGHT	USE
displayLarge	32px	700	Hero titles
titleLarge	20px	600	Screen titles
titleMedium	16px	600	Section headers
bodyLarge	15px	400	Primary body copy
bodyMedium	14px	400	Secondary text
labelLarge	14px	600	Button labels

14.3 Component Specifications

COMPONENT	SPECIFICATION
Cards	16px radius, 1px border, zero elevation (flat design)
Buttons (primary)	Primary colour fill, 52px min height, 14px radius, full-width
Buttons (outline)	1px border, no fill, 14px radius
Input fields	Filled style, 12px radius, 16px horizontal padding; focus: 1.5px primary border
Chips	20px radius, bordered, rarity colours for badges
FAB (extended)	Primary fill, black text/icon, "Add {Category}" label
Bottom Nav	Fixed, no elevation, primary icon colour on active
Tab Bars	Scrollable, label-size indicator, 2.5px indicator weight

15. Go-to-Market Strategy

15.1 Target Audiences

Primary: Domestic Travellers (25–45)

- Indian tourists planning trips to Northeast India.

- Digital-native, comfortable with app-based discovery.
- Seeking authentic, off-the-beaten-path experiences.
- Value local recommendations over generic travel aggregators.

Secondary: International Travellers

- Adventure tourism enthusiasts.
- Photography/nature travel community.
- Cultural heritage explorers.
- Budget backpackers and sustainable travel advocates.

Tertiary: Local Content Creators & Contributors

- Mizoram residents who are proud of their state.
- Local bloggers, photographers, food reviewers.
- Students and young professionals.
- Incentivised by XP, badges, leaderboard visibility, and future MezoPerks rewards.

Quaternary: Tour Operators & Local Businesses

- Small and medium tour operators in Mizoram.
- Homestay owners, adventure guides, cultural experience providers.
- Motivated by free digital storefront and direct booking management.

15.2 Launch Phase Strategy

Phase 1 — Mizoram Beta Launch (V1)

- Soft launch with Mizoram content.
- Seed database via admin CSV bulk upload of existing tourist data.
- Partner with 5–10 established tour operators for venture listings.
- Influencer seeding: partner with Mizo travel Instagram and YouTube creators.
- Campus ambassador programme at Mizoram University, ICFAI, and NIT Mizoram.
- Target: **1,000 registered users in first 60 days**.
- Target: **50 community photo contributions in first 30 days**.

Phase 2 — Community Activation

- Dare campaign: "Explore 5 Hidden Gems of Mizoram" — first-ever platform-wide dare with prizes.
- Review drive: "Rate Your Favourite Spot" — multiplied XP week for reviews.
- Partner with Mizoram Tourism Department for official endorsement.
- Target: **5,000 users, 500 reviews, 20 active tour operators** by end of Phase 2.

Phase 3 — Northeast Expansion

- Activate Nagaland (Hornbill Festival tie-in), Meghalaya (Living Root Bridges), Manipur.
- Multi-state leaderboard seasons.
- Cross-state Dare events.
- Target: **25,000 users across 3 states**.

15.3 Monetisation Roadmap (Post-V1)

STREAM	MECHANISM	TIMELINE
Operator Commission	5–10% on confirmed venture bookings	Phase 2
Featured Listings	Pay-to-be-featured — promoted spots on home screen	Phase 2
MezoPerks Loyalty	Partnerships with local businesses for XP reward redemption	Phase 3
Premium Users	Remove dare/bucket list cap; exclusive badges; advanced analytics	Phase 2
Sponsored Dares	Brands sponsor platform-wide travel challenges	Phase 3
AI Planner (Premium)	AI itinerary generation subscription	Phase 4

15.4 Distribution Channels

- **App Stores**: iOS App Store + Google Play Store (primary).
- **WhatsApp**: Organic sharing via `share_plus` (spot cards, venture links).
- **Social Media**: Instagram (visual tourism content), Facebook (Mizoram community groups), YouTube (operator showcase videos).
- **Government Partners**: Mizoram Tourism Department official partnership for credibility and data access.
- **Travel Agencies**: B2B referral partnerships for venture bookings.
- **University Networks**: Student ambassador programme for contributor acquisition.

15.5 Retention Mechanics

MECHANIC	PURPOSE
Daily Login Streak	Habit formation; escalating XP multiplier punishes streak breaks
Badge Notifications	Milestone-triggered push to re-engage dormant users
Weekly Leaderboard Reset	Creates urgency for weekly engagement spikes
New Dare Notifications	Social pressure from friend group activities

MECHANIC	PURPOSE
Booking Status Updates	Keeps users in the booking funnel until completion
Review Follow-up	Post-visit prompt to submit experience rating

---

16. Innovation & Vision

16.1 What Makes Xplooria Different

1. Hyper-Local Focus

Unlike global platforms (TripAdvisor, Google Maps), Xplooria is built exclusively for Northeast India. Every feature, category, and data model is designed for the specific travel patterns, geography, and culture of the region. This hyper-local focus enables:

- Categories that matter locally (Homestays, Adventure Spots, Cultural Sites) — not generic hotel chains.
- Local operator onboarding with no technical overhead.
- Content verified by people who actually know the region.

2. Gamification as Core Infrastructure (Not an Afterthought)

Gamification is not a "layer" on top of the product — it is embedded in the Firestore transaction layer. Every meaningful user action triggers an atomic XP award that:

- Cannot be double-awarded (transaction integrity).
- Cannot be lost (even if the UI crashes, the XP write persists).
- Cascades to badge evaluation, level recalculation, and streak tracking simultaneously.

3. Social Travel Planning — Beyond Lists

The combination of Bucket Lists (collaborative destination lists), Dares (proof-based group challenges), and Dilemmas (community voting) creates a full social travel planning ecosystem that no existing travel app offers in the Indian market.

4. Operator Empowerment

Local tour operators get:

- A mobile storefront with full package/pricing configuration.
- Direct booking management with status tracking.
- Customer feedback collection.
- WhatsApp integration (operator contact via WhatsApp is standard in India).
- All at zero cost in V1.

5. Resilient Offline-First Architecture

`flutter_secure_storage` and `shared_preferences` cache critical data locally.

`connectivity_plus` monitors network state. Future offline database (Hive or SQLite) is already in the architecture roadmap.

6. AI Travelling Planner (Roadmap)
The V1 UI already includes a locked "Coming Soon" teaser for the AI Travelling Planner & Companion feature. This teaser:
- Communicates the platform's AI ambition.
- Sets user expectations for the next release.
- Validates interest before building (lean startup signal).

Planned AI capabilities:
- **Smart Itinerary Generation** — Input: start date, interests, budget → Output: day-by-day itinerary with Xplooria listings.
- **AI Companion Chat** — Answers travel questions contextually using Xplooria's curated data.
- **Personalised Recommendations** — ML-powered spot suggestions based on review history and browsing patterns.
- **Live Suggestions** — Real-time context-aware tips based on GPS location.

16.2 Technical Innovation

Atomic Gamification Transactions
The `GamificationService.award()` method uses Firestore's multi-document transaction to ensure XP, badge evaluation, level recalculation, streak tracking, and audit logging happen atomically — no partial state possible.

Dual-Shell Admin Architecture
The admin and consumer apps share the same codebase but are separated by GoRouter's `ShellRoute` with independent `HeroControllerScope`. This eliminates the need for two separate Flutter projects while maintaining complete UI/UX isolation between admin and consumer modes.

Real-time Earned-State Architecture
Profile data (XP, level, badges) updates in real-time for the user who triggered the action via `authController.refreshProfile()`, and for any other viewer of their profile via `currentUserStreamProvider` (Firestore live stream). No polling required.

Security-by-Design
Both client-side (GoRouter) and server-side (Firestore rules) enforce the same role model independently. Even if the GoRouter is bypassed, Firestore rules prevent unauthorised writes.

16.3 Long-Term Vision (3-Year Roadmap)

MILESTONE	DESCRIPTION
V1 (Current)	Mizoram, all core features, admin panel, gamification

MILESTONE	DESCRIPTION
V1.5	Push notifications (FCM), offline reading mode, enhanced search
V2	AI Travelling Planner (basic), 3 new NE states
V2.5	MezoPerks loyalty programme live, operator analytics dashboard
V3	Full NE coverage (all 8 states), AI companion, premium subscriptions
V4	Pan-India tribal/rural tourism expansion, B2B API for travel agencies

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17. Validation & Credibility

17.1 Technical Credibility

Production-Grade Stack

- Firebase (used by Google, Alibaba, The New York Times) — enterprise reliability.
- Flutter (used by Google Pay, BMW, Alibaba) — proven performance at scale.
- Riverpod (the successor to Provider, endorsed by the Flutter team) — state management best practices.

Security Compliance

- All user passwords handled exclusively by Firebase Auth (never stored in Firestore or Storage).
- Role-based access control enforced at the database level via Firestore Security Rules.
- AWS-equivalent SLA via Firebase infrastructure (99.95% uptime guarantee).
- Image uploads size-limited and MIME-type validated in Storage Rules.
- No PII exposed in Firestore indexes — email addresses are protected by `isUser(uid)` rules.

Code Quality Signals

- Immutable data models via Freezed (prevents accidental state mutation).
- Sealed result types (`ApiResult`, `AuthResult`) prevent unhandled null states.
- `autoDispose` providers for admin collections — no memory leaks from unused data listeners.
- CI-ready: `flutter_lints` enforced, test infrastructure with `fake_cloud_firestore` and `firebase_auth_mocks`.

17.2 Market Validation

TAM (Total Addressable Market)

- 2.5 million+ domestic tourists travel to Northeast India annually (Ministry of Tourism, 2023).

- International tourist arrivals growing at ~12% YoY post-COVID.
- 780+ million smartphone users in India; 50 million+ active travel app users.

SAM (Serviceable Addressable Market) — V1
- Mizoram population: ~1.3 million.
- Estimated smartphone penetration: ~60% = 780,000 smartphone users.
- Travel-interested segment: ~15% = ~120,000 potential users.

SOM (Serviceable Obtainable Market) — Year 1
- Target: **10,000 registered users** in Mizoram by end of Year 1.
- Target conversion rate from downloads to registrations: 35%.
- Operator acquisition target: **50 active venture listings**.

Competitive Gap

Platform	NE India Coverage	Local Tours	Gamification	Community Planning
Google Maps	Partial	No	No	No
TripAdvisor	Minimal	No	Badges only	No
MakeMyTrip	No tours	Hotel/flight only	No	No
Xplooria	**Full (Mizoram first)**	**Yes**	**Full system**	**Yes**

No direct competitor addresses the Northeast India market with the depth that Xplooria provides.

17.3 Partnership Signals

- **Mizoram Tourism Department** — Official data partnership opportunity (spot data, event data).
- **Local Tour Operators** — Free platform = zero adoption friction; validated interest from regional operators.
- **University Partnerships** — Campus ambassador readiness at Mizoram University.
- **HillsTech** — Local technology company with regional credibility and relationships.

17.4 Platform Trust Features

- **One review per user per place** — Prevents review bombing and fake ratings.
- **Admin moderation queue** — All user contributions require admin approval before going live.
- **Contribution XP only on approval** — No incentive to spam contributions (reward only after admin verification).
- **Booking status trail** — Transparent booking lifecycle with admin notes prevents dispute ambiguity.
- **Visitor Guide Dos/Don'ts** — Cultural sensitivity built into the product (key for Northeast India tourism).

---

18. Version 1 Feature Completeness Matrix

FEATURE CATEGORY	FEATURE	STATUS
Authentication	Email/password register	✅ Complete
	Email/password login	✅ Complete
	Password reset	✅ Complete
	Auth state persistence	✅ Complete
	Role-based redirect	✅ Complete
Discovery	Tourist spots (browse + detail)	✅ Complete
	Restaurants	✅ Complete
	Cafes	✅ Complete
	Hotels / Stay	✅ Complete
	Homestays	✅ Complete
	Adventure spots	✅ Complete
	Shopping areas	✅ Complete
	Events	✅ Complete
	Ventures / Tour packages	✅ Complete
	Global search	✅ Complete
Maps	Interactive community map	✅ Complete
	Category filtering (icon chips)	✅ Complete
	Animated map pins	✅ Complete
	Tap-to-detail	✅ Complete
Community	Feed posts (text + image)	✅ Complete

FEATURE CATEGORY	FEATURE	STATUS
	Like / comment	✅ Complete
	Dilemma A/B polls	✅ Complete
	Bucket lists (rooms)	✅ Complete
	Dares (group challenges)	✅ Complete
	Dare photo proof	✅ Complete
	Dare scratch cards	✅ Complete
	Notifications	✅ Complete
Gamification	XP system (11 actions)	✅ Complete
	10-level progression	✅ Complete
	22 badges	✅ Complete
	Daily login streak + multiplier	✅ Complete
	Leaderboard	✅ Complete
	XP toast notification	✅ Complete
Bookings	Browse ventures	✅ Complete
	Select package + add-ons	✅ Complete
	Booking review + submit	✅ Complete
	"My Bookings" tracking	✅ Complete
	Post-booking feedback	✅ Complete
Profile	View stats and badges	✅ Complete
	Edit profile (name, bio, photo)	✅ Complete
	XP history feed	✅ Complete
	Theme toggle (dark/light)	✅ Complete

FEATURE CATEGORY	FEATURE	STATUS
Home	Banner carousel (Firestore-driven)	✅ Complete
	Category grid shortcuts	✅ Complete
	Featured spots section	✅ Complete
	Tour packages preview	✅ Complete
	Visitor Guide card	✅ Complete
	NE State picker	✅ Complete
	AI Planner teaser (locked)	✅ Complete
Visitor Guide	State guide (Mizoram)	✅ Complete
	Dos / Don'ts	✅ Complete
	Quick facts	✅ Complete
Admin — Content	Manage 8 collection types	✅ Complete
	Add/Edit/Delete listings	✅ Complete
	CSV bulk import	✅ Complete
	Manage ventures (12-section form)	✅ Complete
	Banner management	✅ Complete
	Visitor guide management	✅ Complete
Admin — Operations	View all bookings	✅ Complete
	Update booking status	✅ Complete
	View venture feedback	✅ Complete
	User management	✅ Complete
	Analytics dashboard	✅ Complete
	Moderation queue	✅ Complete

FEATURE CATEGORY	FEATURE	STATUS
Upcoming (Roadmap)	AI Travelling Planner	🔒 Coming Soon
	Push notifications (FCM)	🔲 Planned
	MezoPerks loyalty redemption	🔒 Coming Soon
	Offline reading mode	🔲 Planned
	Multi-state expansion	🔲 Planned

19. Appendix A: Complete Route Map

ROUTE	PATH	AUTH REQUIRED	ADMIN ONLY
Onboarding	`/onboarding`	No	—
Login	`/login`	No	—
Register	`/register`	No	—
Forgot Password	`/forgot-password`	No	—
Home	`/`	No	—
Spots List	`/spots`	No	—
Spot Detail	`/spots/:id`	No	—
Listings (tab)	`/listings?tab={n}`	No	—
Restaurant Detail	`/listings/restaurants/:id`	No	—
Hotel Detail	`/listings/hotels/:id`	No	—

ROUTE	PATH	AUTH REQUIRED	ADMIN ONLY
Cafe Detail	`/listings/cafes/:id`	No	—
Stay Detail	`/listings/stay/:id`	No	—
Adventure Detail	`/listings/adventure/:id`	No	—
Shopping Detail	`/listings/shopping/:id`	No	—
Search	`/search`	No	—
Community (Feed/Map)	`/community`	No	—
Create Post	`/community/new`	**Yes**	—
Create Dilemma	`/community/dilemmas/new`	**Yes**	—
Create Bucket List	`/community/bucket-lists/new`	**Yes**	—
Bucket List Detail	`/community/bucket-lists/:id`	**Yes**	—
Edit Bucket List	`/community/bucket-lists/:id/edit`	**Yes**	—
Add Bucket Item	`/community/bucket-lists/:listId/add-item`	**Yes**	—
My Rooms	`/community/my-rooms`	**Yes**	—
Create Dare	`/community/dares/new`	**Yes**	—
Dare Detail	`/community/dares/:id`	**Yes**	—
Edit Dare	`/community/dares/:id/edit`	**Yes**	—

ROUTE	PATH	AUTH REQUIRED	ADMIN ONLY
Add Challenge	`/community/dares/:dareId/add-challenge`	**Yes**	—
Submit Proof	`/community/dares/:dareId/challenges/:challengeId/proof`	**Yes**	—
Dare Rewards	`/dare-rewards`	**Yes**	—
Scratch Card	`/dare-rewards/cards/:cardId`	**Yes**	—
Notifications	`/notifications`	**Yes**	—
Dare Dashboard	`/profile/dare-dashboard`	**Yes**	—
Venture Detail	`/ventures/:id`	No	—
Booking Review	`/ventures/:id/booking-review`	**Yes**	—
My Bookings	`/my-bookings`	**Yes**	—
Event Detail	`/events/:id`	No	—
Reviews	`/reviews/:collection/:id`	No	—
Leaderboard	`/leaderboard`	No	—
Profile	`/profile`	**Yes**	—
Edit Profile	`/profile/edit`	**Yes**	—
Admin Shell	`/admin`	**Yes**	**Yes**
Admin Listings	`/admin` (tab)	**Yes**	**Yes**

ROUTE	PATH	AUTH REQUIRED	ADMIN ONLY
Admin Add Listing	`/admin/listings/add/:collection`	**Yes**	**Yes**
Admin Edit Listing	`/admin/listings/edit/:collection/:docId`	**Yes**	**Yes**
Admin Add Venture	`/admin/ventures/add`	**Yes**	**Yes**
Admin Edit Venture	`/admin/ventures/edit/:docId`	**Yes**	**Yes**

20. Appendix B: Dependencies Reference

Production Dependencies Summary

CATEGORY	COUNT	KEY PACKAGES
Core Framework	2	flutter, flutter_riverpod
Firebase	5	firebase_core, auth, firestore, storage, analytics
Navigation	1	go_router
Networking	2	dio, pretty_dio_logger
State/Serialization	4	riverpod_annotation, freezed_annotation, json_annotation, equatable
Local Storage	2	shared_preferences, flutter_secure_storage
UI Components	10	carousel_slider, iconsax, google_fonts, cached_network_image, shimmer, lottie, fl_chart, confetti, percent_indicator, smooth_page_indicator
Maps & Location	4	flutter_map, latlong2, geolocator, geocoding

CATEGORY	COUNT	KEY PACKAGES
Forms & Validation	2	flutter_form_builder, form_builder_validators
Media	3	image_picker, image_cropper, image
Bulk Upload	2	file_picker, csv
Utilities	8	intl, timeago, uuid, url_launcher, share_plus, connectivity_plus, package_info_plus, logger
Functional	1	dartz

Total Production Dependencies: 46+

Development Dependencies

PACKAGE	PURPOSE
build_runner	Code generation orchestrator
freezed	Immutable model generation
json_serializable	JSON codec generation
riverpod_generator	Provider code generation
flutter_lints	Lint rules
flutter_launcher_icons	App icon generation
flutter_test	Unit + widget testing
mockito	Mock generation for tests
fake_cloud_firestore	In-memory Firestore for tests
firebase_auth_mocks	Firebase Auth mock for tests

Document Information

FIELD	VALUE
Document Title	Xplooria V1 Product Documentation
Version	1.0
Date	April 2026
Author	HillsTech Engineering Team
Platform	iOS & Android (Flutter)
Backend	Firebase (spotmizoram project)
Document Location	`/documentation/XPLOORIA_V1_DOCUMENTATION.md`

This document represents the complete feature, architecture, strategy, and technical reference for Xplooria Version 1. For questions regarding specific implementation details, refer to the source code at `lib/` or contact the HillsTech engineering team.